

HARLAN, IA, USA

WMO: 722097

Lat: 41.584N

Lon: 95.339W

Elev: 375

StdP: 96.90

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04936

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-18.0	-25.0	0.4	-21.0	-22.4	0.5	-18.0	14.6	-3.6	12.9	-5.4	2.4	190	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.8	24.0	32.1	23.9	30.3	22.9	26.6	30.5	25.4	29.4	24.4	28.5	5.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.3	28.9	24.0	19.8	27.8	22.9	18.4	27.0	85.0	30.7	80.1	29.3	75.7	28.7	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.8	9.3	DB	-26.5	35.1	2.8	1.0	-28.5	35.8	-30.2	36.3	-31.8	36.9	-33.8	37.6	(4)
(5)			WB	-26.7	28.3	2.8	1.1	-28.7	29.1	-30.4	29.7	-32.0	30.3	-34.0	31.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-5.5	-4.1	4.0	9.9	16.7	21.9	23.3	21.6	18.5	11.3	3.7	-3.3	(6)	
	DBStd	11.58	6.78	6.86	6.76	5.60	5.00	3.27	3.21	3.21	4.35	5.35	5.79	6.30	(7)	
	HDD10.0	1823	482	396	208	69	6	0	0	0	1	48	200	413	(8)	
	HDD18.3	3568	740	629	444	257	91	7	4	9	50	227	438	672	(9)	
	CDD10.0	1789	0	0	23	66	214	356	412	361	258	88	12	1	(10)	
	CDD18.3	493	0	0	1	4	41	113	158	111	57	8	0	0	(11)	
	CDH23.3	4894	0	0	11	78	451	1134	1588	970	556	104	3	0	(12)	
	CDH26.7	1674	0	0	2	20	145	407	601	300	171	28	0	0	(13)	
(14) Wind	WSAvg	3.9	4.3	4.2	4.7	5.1	4.6	3.7	2.6	2.4	3.1	3.7	4.2	4.0	(14)	
(15) Precipitation	PrecAvg	847	23	26	59	90	129	110	111	97	76	57	41	26	(15)	
	PrecMax	1150	48	51	132	198	241	250	212	215	219	158	85	65	(16)	
	PrecMin	576	8	3	2	17	37	21	36	9	21	12	2	0	(17)	
	PrecStd	155	10	15	36	42	64	61	54	54	49	34	24	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	16.4	24.8	29.2	33.7	34.0	34.8	33.0	32.8	30.1	22.5	13.7	(19)	
		MCWB	6.0	8.6	14.6	17.0	20.1	22.8	25.4	24.5	23.0	20.5	13.8	9.7	(20)	
	2%	DB	7.8	12.1	21.3	25.3	30.2	32.4	32.7	32.0	31.1	26.2	18.7	10.9	(21)	
		MCWB	4.6	7.3	13.5	14.6	19.9	22.7	25.4	24.8	22.4	17.4	13.0	7.8	(22)	
	5%	DB	5.2	7.7	17.7	22.4	27.6	31.0	32.0	30.0	28.7	22.9	16.0	7.6	(23)	
		MCWB	3.0	4.7	12.1	13.7	18.2	22.0	25.2	24.2	21.8	15.5	11.1	5.1	(24)	
	10%	DB	2.8	5.0	14.8	19.9	26.0	28.9	30.1	27.8	27.1	21.0	12.7	5.1	(25)	
		MCWB	1.3	3.0	10.4	13.0	17.5	21.3	24.3	22.9	20.1	14.4	8.3	3.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	11.8	17.2	18.7	23.0	26.5	28.5	27.3	25.1	22.8	16.0	12.0	(27)	
		MCDB	9.8	16.0	22.0	24.2	28.3	30.5	31.5	31.5	29.8	27.5	18.2	12.9	(28)	
	2%	WB	5.1	7.6	15.4	16.7	21.5	24.8	27.1	26.1	23.9	19.9	13.7	7.7	(29)	
		MCDB	8.0	10.7	19.1	22.3	27.0	29.2	30.6	30.0	28.5	23.6	17.9	8.7	(30)	
	5%	WB	3.2	5.0	13.0	15.4	20.4	23.8	25.8	25.0	22.9	17.5	11.4	5.2	(31)	
		MCDB	5.2	7.6	16.6	20.7	25.7	28.6	29.8	28.7	27.0	20.3	15.3	7.8	(32)	
	10%	WB	1.7	3.0	10.3	13.9	19.1	22.7	24.8	23.9	21.8	15.2	9.0	3.2	(33)	
		MCDB	3.1	5.0	15.4	18.8	23.7	27.4	28.9	27.3	25.4	19.3	12.9	5.1	(34)	
(35) Mean Daily Temperature Range	MDBR		9.7	10.2	11.5	12.7	12.1	11.5	11.4	11.2	13.1	12.7	11.7	9.6	(35)	
	5% DB	MCDBR	12.7	13.7	16.7	17.2	14.7	13.3	12.5	12.7	14.1	16.2	16.0	13.0	(36)	
		MCWBR	9.5	9.7	9.6	8.8	6.2	5.9	6.4	6.9	6.7	8.3	9.6	9.8	(37)	
	5% WB	MCDBR	11.8	13.2	14.0	13.8	11.9	11.3	10.6	11.0	11.5	12.1	14.4	11.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.281	0.310	0.326	0.380	0.403	0.411	0.427	0.413	0.376	0.332	0.295	0.285	(40)	
	taud		2.429	2.365	2.396	2.265	2.271	2.305	2.248	2.298	2.357	2.466	2.498	2.440	(41)	
	Ebn at Noon		877	897	923	889	874	865	846	847	857	862	855	840	(42)	
	Edh at Noon		77	97	106	130	133	128	135	125	109	87	72	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.95	2.93	3.78	4.81	5.74	6.36	6.45	5.55	4.60	3.17	2.18	1.64	(44)	
	RadStd		0.18	0.18	0.35	0.36	0.43	0.53	0.43	0.38	0.30	0.38	0.15	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page